

Research Plan — QIntern 2026

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1. Project Title

Heterogeneity-Aware QUBO Formulation for Task Mapping in Network-on-Chip Architectures via Quantum Annealing

2. Project Overview

Network-on-Chip (NoC) architectures are widely used as scalable communication fabrics for many-core system-on-chip (SoC), accelerator, and chiplet-oriented designs. As these systems become increasingly heterogeneous, routers may differ in micro-architecture, links may vary in bandwidth or operating frequency, and three-dimensional stacking may introduce through-silicon vias (TSVs) with distinct latency and energy profiles. The central design-time problem considered in this project is task mapping: given an application modeled as a task communication graph and a weighted NoC topology, assign each task to a processing tile so as to minimize inter-tile communication cost, energy consumption, and peak latency.

Task mapping is NP-hard in the general case. Classical optimization methods, including branch-and-bound ILP, simulated annealing, genetic algorithms, and NoC-specific heuristics such as NMAP, have been extensively studied for related mapping problems [1, 2]. However, the increasing heterogeneity of emerging NoC, 3D-NoC, and chiplet-based systems motivates new formulations and benchmarking studies. Recent literature indicates that quantum annealing and Quadratic Unconstrained Binary optimization (QUBO) models have been explored for related combinatorial assignment, scheduling, placement, and routing problems, but direct QUBO/quantum-annealing studies for heterogeneous NoC task mapping remain limited. This project addresses that gap as a formulation and benchmarking study.

Let $\mathcal{T} = \{1, \dots, N\}$ be the task set, $\mathcal{P} = \{1, \dots, K\}$ the NoC tile set, and $\mathcal{E} \subseteq \mathcal{T} \times \mathcal{T}$ the communicating task pairs with communication volume $v_{ij} \geq 0$. Let $d_{kl} \geq 0$ denote the heterogeneous routing cost between tiles k and l , precomputed by all-pairs shortest paths on the weighted topology graph. Binary variables $x_{ik} \in \{0, 1\}$ encode the assignment ($x_{ik} = 1$ iff task i is mapped to tile k). The core quadratic objective is

$$\min_{\mathbf{x}} \sum_{(i,j) \in \mathcal{E}} \sum_{k,l \in \mathcal{P}} v_{ij} d_{kl} x_{ik} x_{jl}. \quad (1)$$

Each task must be mapped to exactly one tile, enforced by the standard one-hot penalty

$$\lambda_1 \sum_{i \in \mathcal{T}} \left(\sum_{k \in \mathcal{P}} x_{ik} - 1 \right)^2. \quad (2)$$

For the one-task-per-tile case, tile exclusivity can be encoded by the pairwise quadratic penalty

$$\lambda_2 \sum_{k \in \mathcal{P}} \sum_{i < j} x_{ik} x_{jk}. \quad (3)$$

The formulation will also include a memory-capacity constraint for each tile. Let m_i be the memory requirement of task i and M_k the memory capacity of tile k . A valid mapping must satisfy

$$\sum_{i \in \mathcal{T}} m_i x_{ik} \leq M_k, \quad \forall k \in \mathcal{P}. \quad (4)$$

This inequality will be handled either through D-Wave’s Constrained Quadratic Model (CQM) representation or through a BQM/QUBO conversion with binary slack variables.

The project’s contributions are threefold:

1. A **heterogeneity-aware QUBO/CQM formulation** that incorporates non-uniform routing costs induced by heterogeneous links, routers, and TSVs.
2. A **distance-precomputation pipeline** that converts weighted topology information into a tile-pair cost matrix compatible with the quadratic mapping objective.
3. A **benchmarking study** comparing D-Wave hybrid solvers, BQM penalty-encoded quantum annealing for small instances, and classical baselines such as simulated annealing, NMAP, and ILP on identical task graph instances.

3. Expected Background and Preparation

Interns are expected to have:

- Basic understanding of quantum computing concepts such as qubits, gates, and measurement.
- Completion of the QCobalt workshop and familiarity with combinatorial optimization concepts such as binary programming, ILP, and simulated annealing.
- Working knowledge of Python; prior exposure to `numpy`, `scipy`, and `networkx` is helpful.

As preparation, interns are encouraged to: (i) complete the D-Wave Ocean SDK getting-started tutorial (https://docs.dwavequantum.com/en/latest/ocean/index_get_started.html); (ii) study the QUBO tutorial of Glover et al. [6] and the Ising formulation overview by Lucas [7]; and (iii) read the background sections of Hu and Marculescu [1] and Murali and De Micheli [2] for NoC mapping context.

4. Execution Plan

Week	Milestones
1	Onboarding; D-Wave Leap/Ocean setup; focused literature review; NoC topology modelling; selection of synthetic and real task graph benchmarks.
2	Heterogeneous distance-precomputation pipeline using <code>networkx</code> ; implementation or adaptation of classical baselines, including simulated annealing and NMAP-style mapping.
3	QUBO/CQM formulation; penalty and capacity-handling experiments; validation on small topologies against exhaustive search or ILP.
4	D-Wave hybrid CQM experiments; BQM/QPU feasibility tests on small instances; embedding quality and chain-break analysis where applicable.
5	Benchmarking on synthetic and application-inspired task graphs; heterogeneity ablation study (homogeneous mesh, variable-link mesh, and small 3D-NoC examples); sensitivity analysis for penalty weights.
6	Result analysis; code/documentation cleanup; final report and QIntern mini-workshop presentation preparation.

5. Deliverables

- **Open-source Python prototype** implementing the heterogeneous distance pipeline, QUBO/CQM construction, classical baselines, and D-Wave solver interface.
- **Reproducible benchmark instances** including synthetic random task graphs and selected application-inspired task graphs such as MPEG, H.264, VOPD, or MWD, subject to availability and clean redistribution conditions.
- **Final technical report and workshop presentation** for the QIntern 2026 closing event.
- **Target post-program output:** an arXiv preprint or short workshop/conference manuscript, depending on the strength of the experimental results.

6. Guidance and Mentor Background

The mentor will be actively involved in the theoretical and implementation aspects of the project through weekly video meetings and asynchronous support. Additional meetings can be arranged when needed. The mentor’s research provides direct methodological grounding [3, 4, 5], and establishes prior work on NoC-based mapping and communication-aware optimization. The project is suitable for interns with backgrounds or interest in combinatorial optimization, embedded systems, computer architecture, or introductory quantum computing.

Risk mitigation. The project is explicitly framed as a formulation and benchmarking study, not as a quantum-advantage claim. Direct QPU runs will be restricted to small instances within embedding feasibility. Larger instances will use D-Wave hybrid solvers and classical baselines. If quantum hardware access is limited, the project can still be completed using Ocean-compatible simulated annealing, exact/ILP validation for small instances, and hybrid solver runs where available.

References

- [1] J. Hu and R. Marculescu, “Energy-aware mapping for tile-based NoC architectures under performance constraints,” in *Proceedings of the ASP-DAC Asia and South Pacific Design Automation Conference (ASP-DAC)*, Kitakyushu, Japan, 2003, pp. 233–239, doi: 10.1109/ASPDAC.2003.1195022.
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- [3] F. Nacar, A. Cakin, S. Dilek, S. Tosun, and K. Chakrabarty, “Neuron grouping and mapping methods for 2D-mesh NoC-based DNN accelerators,” *Journal of Parallel and Distributed Computing*, vol. 193, Art. 104949, 2024, doi: 10.1016/j.jpdc.2024.104949.
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- [5] A. Cakin, S. Dilek, and S. Tosun, “An enhanced integer linear programming model for communication-aware and load-balanced neuron grouping in manycore neural network accelerators,” *Concurrency and Computation: Practice and Experience*, vol. 38, no. 7, Art. e70676, 2026, doi: 10.1002/cpe.70676.
- [6] F. Glover, G. Kochenberger, R. Hennig, and Y. Du, “Quantum bridge analytics I: a tutorial on formulating and using QUBO models,” *Annals of Operations Research*, vol. 314, pp. 141–183, 2022, doi: 10.1007/s10479-022-04634-2.
- [7] A. Lucas, “Ising formulations of many NP problems,” *Frontiers in Physics*, vol. 2, Art. 5, 2014, doi: 10.3389/fphy.2014.00005.